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Beyond Short-Term Noise: Robust Long-Horizon Bankruptcy Prediction Using Gated Dual-Branch Transformers

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ABSTRACT

Long-term corporate solvency forecasting is fundamentally constrained by two intertwined obstacles: severe class imbalance in default events and the nonstationary dynamics of financial markets. Deep learning architectures optimized on short-horizon objectives characteristically overfit to transient market noise, yielding predictive performance that deteriorates sharply as the forecasting horizon extends. This instability is addressed through the Gated Spectral–Temporal Transformer (GST-Net) a hybrid architecture that introduces a learnable gating mechanism to suppress transient, high-frequency signal components in favor of persistent structural trends. A dual-branch design further decouples the modeling of localized anomalies from global temporal dependencies, enabling the network to preserve discriminative signal across multiyear horizons where conventional models degrade. Given the extreme class imbalance inherent to solvency prediction, the evaluation protocol centers on the Matthews correlation coefficient and macro F1 score, metrics that penalize minority-class misclassification far more rigorously than accuracy alone. At a three-year forecasting horizon, GST-Net achieves an MCC of 0.5704, a macro F1 score of 78.38%, and an accuracy of 95.95%, nearly doubling the MCC of the strongest tuned baseline, XGBoost, which peaked at 0.3294, while state-of-the-art tabular deep learning models, TabNet and TabPFN, collapse entirely under identical conditions. Beyond predictive robustness, GST-Net sustains subsecond inference latency of approximately 0.37 s per instance, rendering it practical for quarterly operational auditing at scale. Coupling the model with SHAP and LIME yields the interpretability required for regulatory scrutiny, exposing an economically coherent shift from short-term liquidity stress to long-term structural decay as the dominant predictive signal, an insight with direct implications for early-warning credit risk systems.

1 | Introduction

Economic and financial stability is one of the essential parts of the long-term economic resilience. Bankruptcies within a corporation remain to cause wide-reaching upheavals that formidably impact international markets. Early detection of firms that are at risk is one of the main interests of modern academic

research [1, 2]. Financial ratios and expert judgment, which are primitive, have long been used by analysts. Nonetheless, the growing complexity of data presents serious restrictions to these manual methods [3].

Recently, there has been a rise in innovative approaches to predictive modeling owing to the recent developments in

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machine learning [4, 5]. Structured accounting information is still at the center stage of accurate corporate financial conjecture, and unstructured textual revelation contains abundant indications of the health of a firm operation [6, 7]. The combination of these distinct data modalities presents a significant technical problem. So far, the particular problem has been a relatively unexplored field in terms of research related to bankruptcy prediction [8, 9].

The current paper proposes a hybrid model, which is known as Gated Spectral–Temporal Transformer (GST-Net), to predict bankruptcy. The architecture suggested is a combination of convolutional layers and high-end transformer encoder blocks. The mechanism is a feature-gating mechanism that dynamically suppresses noisy signals of input sequences. Both the local spectral volatility and global temporal trends are captured by parallel branches. A skewed loss is used to deal with extreme imbalance of classes during training. Ensemble-based learning methods minimize the differences in prediction and improve the general model strength. Empirical assessments reveal that GST-Net has the best predictive value of long-term company bankruptcies. The experimental findings indicate that GST-Net is superior to the existing deep learning baselines. Explainable modules like SHAP provide clear explanations to financial decision makers. This integrative model is an effective way of responding to limitations of existing financial analysis. The superior performance is confirmed by strong statistical analysis in more than 5 years of independent testing. Rather than proposing fundamentally novel atomic neural operations, the core contribution of this research resides in a highly rigorous applied engineering synthesis. By carefully adapting, tuning, and structurally integrating established spatial–temporal components—specifically, GLU-inspired gating, parallel CNN-transformer processing, and focal-style asymmetric loss—this framework uniquely addresses the niche and highly complex problem of long-horizon financial ratio stabilization under conditions of extreme class imbalance and systemic market noise. In short, the main contributions of the study are as follows:

1. It suppresses redundant noise and stabilizes learning by employing a learnable gating mechanism alongside asymmetric loss functions to address extreme data imbalance in financial time series.
2. It jointly models local and global temporal dynamics by integrating spectral feature extraction with long-range temporal dependency learning, supported by quantile-based normalization for heavy-tailed distributions.
3. It enhances forecast robustness and long-horizon stability through ensemble learning that averages predictions from multiple independently trained models across one- to three-year horizons.
4. It achieves superior performance with regulatory-ready interpretability by outperforming existing benchmarks while providing firm-level explanations using tools such as LIME.

To facilitate a rigorous and methodical exposition, the remainder of this manuscript is structured into six subsequent sections. Section 2 critically synthesizes the extant literature, establishing the theoretical and methodological context of contemporary bankruptcy prediction models. Section 3 delineates the proposed

GST-Net framework, encompassing initial data preprocessing protocols, the mathematical formulation of the dynamic feature-gating mechanism, and the underlying mechanics of the dual-branch spectral–temporal architecture. Section 4 details the comprehensive suite of evaluation metrics deployed to strictly gauge predictive efficacy, particularly under conditions of extreme class imbalance. Section 5 presents a robust empirical evaluation, offering a nuanced comparative performance analysis against current state-of-the-art (SOTA) baselines. Section 6 reflects upon the salient limitations of the present study while outlining promising trajectories for future investigative endeavors. Finally, Section 7 provides concluding remarks, summarizing the overarching academic and practical contributions of the research. This systematic arrangement ensures a highly coherent and logically progressive analytical narrative.

2 | Related Work

Statistical models that are based on financial ratios are often used as empirical studies in bankruptcy prediction [1, 2]. Initially, the methods were based on linear accounting metrics to outline risky companies. This notwithstanding, complex patterns of data are often not represented by linear rules. Following studies presented new and improved machine learning methods to improve predictive accuracy [10, 11]. These techniques are also skillful in nonlinearity and high dimensional inputs. These functions are also applicable especially when there are subtle interactions of variables that translate to a strong impact [12].

Sequential patterns in financial statement data have been modeled using deep learning methods [13, 14]. Such approaches exploit high-level neural network models to learn hierarchical patterns in the data. Time-series analysis of financial ratios has used recurrent and convolutional neural networks [15, 16], thus revealing dynamic patterns that are typically not revealed by conventional models. Experimentally, it is found that deep learning generally performs better than traditional risk-forecasting models [17], which is most evident in short-term forecasts when large datasets can be accessed. Fasano et al. [18] proposed a new and developed bankruptcy prediction model that combines deep recurrent neural networks (RNNs) with explainable artificial intelligence with the accuracy of 0.85 and area under the receiver operating characteristic curve (ROC–AUC) of 0.927. The research utilized a large pool of data consisting of 4.1 million entries obtained through the Orbis database in the time frame of 2012–2021 with a focus on small- and medium-sized businesses in Europe. The model was used to process variable-length financial time series, improve predictive performance, and identify key financial variables that have high predictive information. As a result, the accuracy of these results was better than the general accuracy of the previous artificial intelligence models.

The textual information is now introduced to the tasks of bankruptcy prediction by researchers [6, 19]. Unstructured data are useful in terms of firm strategy and risk [20]. These lessons are based on disclosures, including disclosures through Managerial Discussion and Analysis (MD&A) sections and disclosures through earnings calls. Initial attempts used unsophisticated bag-of-words models of this textual data [21]. In more recent works, pretrained language models have been used instead of such representations [7, 22]. Text-based models improve prediction

performance when combined with sentiment analysis [23]. Lombardo et al. [24] proposed a transformer-based system for this task. The model combined quantitative accounting information with text excerpts from SEC annual reports from a dataset that included 6190 US public companies that spanned 1999–2018. Its performance scored 87.5% overall. The architecture used a distinct module for the financial and textual inputs, in which the financial figures were modeled separately by RNN and the text was done using a natural language processing summarization technique. Merging of the two modalities took place at a late stage, with the possible drawback of restricting the depth of cross-modal interaction learning. Additionally, the virtually exclusive use of the model with data derived from the public companies of the United States and its reliance on SEC documents limit its use in other parts of the world. Gholampoor et al. [25] used classical machine learning methods, namely, gradient boosting, to forecast the bankruptcy of 1265 healthcare companies by using financial ratios. Their study published more than 90% classification accuracy. However, based on this work, this is limited to only one sector of healthcare and focuses on static financial data collected in 2022; therefore, this work does not include temporal or textual data. Consequently, the methodology is likely to overfit and not generalize well to different industry situations. Moreover, it relies very much on traditional manual feature engineering. Taoushianis et al. [26] used a large firm-level dataset consisting of 1.2 million entities and proposed a non-linear logistic regression model with fractional polynomials, which provides a better fit for complex nonlinear data. Compared to linear models, this dramatically improved in-sample fit, the AUC was increased by eight to 10%, and the Type II error was decreased by 50%. While the model is interpretable, it is only based on the numerical financial ratios, ignoring the unstructured data such as text disclosures and preselected financial variables. Mattos et al. [27] investigated the case of distressed private firms in Brazil by analyzing 503 firms with unreliable financial reporting. They noted that financial ratios are less informative when accounting quality is poor and that institutional and legal variables, such as audit status and amount of collateral, have a greater predictive influence. Their machine learning models were more powerful than traditional statistical methods, but the research lacked providing specificity as to accuracy and has a poor architecture for modeling unstructured text. Idhmad et al. [28] built classifiers with several machine learning models from a Taiwan economic journal dataset between 1999 and 2009. They proposed a combination of financial and corporate governance indicators by using SMOTE to handle more critical class imbalance. Their best performing model achieved an accuracy of 98%. However, the focus on engineered features has the potential to overfit the model, which also does not include any text-based information about corporations, limiting the model's ability to learn from narrative financial disclosures.

Few studies have fully explored multimodal learning for risk prediction [8, 9]. Some work combines numerical and textual features using simple fusion [29, 30]. These strategies often lack proper modality-influence control mechanisms. Recently, transformer-based multimodal frameworks have advanced considerably. These combine heterogeneous sources for financial prediction tasks. Bonnier et al. [31] proposed a tabular-text transformer called *T*. This integrates structured features and

text via cross-modal attention. Their model achieves superior financial risk classification performance. Mathur et al. [32] introduced DocFin, a multimodal financial system. It fuses semistructured table reports and audio data. This enhances prediction accuracy and fairness in forecasts. Cao et al. [33] developed Risk Labs for firm-level risk forecasting. It is based on the use of large language models to combine multisource data, including market indicators and corporate disclosures. Braithwaite et al. [34] present a transformer-based model that unites the sequence of structured transactions and text description. The integration of unstructured textual information alongside quantitative data has significantly advanced the SOTA in bankruptcy prediction over recent years. Seminal studies, such as the work by Mai et al. [35], have demonstrated that deep learning models utilizing textual disclosures capture highly predictive signals regarding corporate health that traditional financial ratios inherently miss. Furthermore, researchers including Nguyen and Huynh et al. [36] have conclusively established that multimodal deep learning frameworks incorporating financial dictionary-based sentiment significantly outperform purely numerical models. While these multimodal approaches define the current predictive frontier, the present study focuses specifically on isolating and resolving the distinct instability and noise issues inherent in extending the forecasting horizon for quantitative sequential data. Their model shows high generalization of financial modeling. These research studies highlight a definite movement to transformer-based architectures.

3 | Methodology

This paper describes a GST-Net in predicting corporate bankruptcy. The sample includes financial data from publicly traded American companies over 2 decades of consecutive financial years. A strict temporal splitting strategy prevents potential look-ahead bias. Sliding windows construct input sequences containing multiple normalized financial accounting ratios. A feature gating mechanism dynamically suppresses noisy or irrelevant features. Two distinct branches process the gated data for feature extraction. Convolutional layers capture spectral patterns while transformers model global temporal dependencies. These multimodal representations are concatenated and fused via linear layers. An AdamW optimizer trains the model using class-aware sampling. The asymmetric loss function effectively handles severe class imbalance issues. Validation scores determine the optimal decision threshold for final testing. An ensemble of three models enhances the final prediction stability. Performance evaluation relies on AUC and correlation metrics. Kernel SHAP analysis provides global feature attribution for model transparency. Local interpretability methods ensure auditability for all relevant stakeholders involved. Figure 1 illustrates the overall architecture of the GST-Net model.

3.1 | Dataset and Preprocessing

3.1.1 | Data Source and Definition

The study utilizes accounting records from major American stock exchanges. The dataset comprises public companies listed on the NYSE and NASDAQ [37]. The temporal scope spans from 1999 to 2018. The collection contains over 8000 distinct firm entities. In total, the dataset includes more than 78,000 firm-year

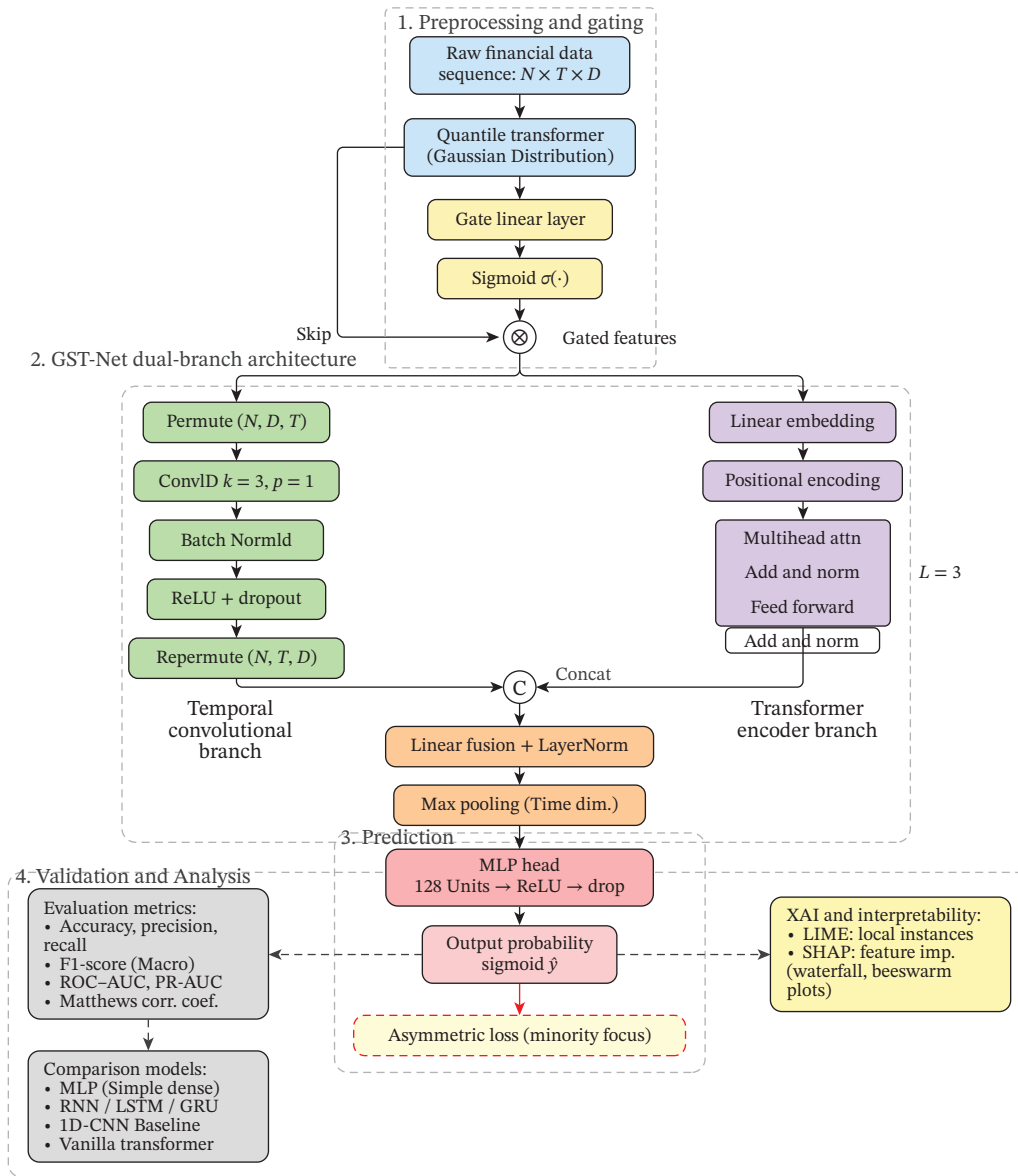


FIGURE 1 | Schematic overview of the proposed gated spectral-temporal transformer (GST-Net) architecture illustrating the feature gating mechanism, dual-branch processing, and classification modules.

observations. Corporate insolvency is strictly defined by legal filing codes that must be officially submitted. Filing for legal protection classifies a firm as bankrupt. Filings under Chapter 7 or Chapter 11 are treated as failure events. The binary target variable represents the state of firm i , where 1 indicates bankruptcy and 0 indicates financial stability. The mathematical formulation of the target variable is given below.

$$y_{i,t} = \begin{cases} 1, & \text{if firm } i \text{ files for Chapter 7 or 11 in year } t, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

3.1.2 | Dataset Characterization and Distributional Dynamics

A rigorous statistical evaluation of the entire dataset encompassing 78,682 firm-year observations across 8971 unique companies reveals distinct structural challenges inherent to corporate financial forecasting. First, the dataset exhibits extreme class

imbalance. Healthy firm years account for 93.37%, corresponding to 73,462 observations, while actual bankruptcy events comprise only 6.63%, corresponding to 5220 observations. This severe skewness fundamentally limits the effectiveness of traditional cross entropy optimization and directly motivates our deployment of an asymmetric loss function to heavily penalize minority class misclassification. Second, a thorough inspection confirmed zero missing values across all financial ratio columns. This high data integrity is a direct artifact of stringent mandatory regulatory reporting standards such as 10-K filings required for publicly traded entities on the NYSE and NASDAQ.

Furthermore, the quantitative features exhibit highly non-stationary and heavy-tailed distributions. As detailed in Table 1, financial ratios display extreme positive skewness reaching up to 29.61 and a substantial prevalence of extreme values. Using the standard interquartile range method with a factor of $1.5 \times \text{IQR}$, we observe that between 13.94% and 22.71% of feature values fall into outlier regions. Figure 2 illustrates this volatility through

TABLE 1 | Statistical characterization of financial features.

Feature	Mean	Std	Skewness	Outlier percentage
X1	880.362485	3928.564794	14.843279	13.948552
X2	1594.529029	8930.484664	20.052194	14.735263
X3	121.234256	652.376804	17.860473	14.584022
X4	376.759424	2012.023142	16.403531	15.528329
X5	201.605717	1060.766096	22.568626	15.804123
X6	129.382453	1265.532022	11.866809	21.611042
X7	286.832743	1335.978571	15.844958	14.281538
X8	3414.354610	18414.103647	18.194283	14.500140
X9	2364.019706	11950.068842	18.987110	14.266287
X10	2867.110620	12917.944421	13.562208	14.642485
X11	722.483710	3242.170946	14.848021	16.065936
X12	255.525035	1494.643534	17.968030	16.841209
X13	769.490783	3774.703114	15.295829	14.369233
X14	610.072255	2938.387443	14.225186	14.972929
X15	532.467069	6369.159440	29.607949	22.707862
X16	2364.019706	11950.068842	18.987110	14.266287
X17	1773.563963	8053.684902	13.767802	15.716428
X18	1987.260307	10419.629038	20.384520	14.252307

logarithmic boxplots. Under these conditions, standard normalization techniques such as Min Max scaling or Z score normalization would severely compress the inlier feature space and distort neural network gradient signals. Consequently, this statistical profile strictly necessitates the use of a nonlinear quantile transformer. This transformation maps heavy-tailed and outlier-affected financial distributions into a uniform Gaussian space, effectively reducing the impact of transient anomalies before signals are processed by the GST-Net.

Value 1 is when bankruptcy filing took place during year t . A value of zero implies a good financial situation at that time. Data partitioning is done on a time basis to avoid look-ahead bias. The training set contains the period between 1999 and 2011. Validation data, which are reserved for hyperparameter tuning and early stopping of the system learning, are between 2012 and 2014. Out-of-sample evaluation uses data from 2015 and later years. The model makes an accurate prediction of insolvency events up to h years in advance where input data are ending in year t , making a prediction for the future status.

3.1.3 | Temporal Sequence Construction

Data partitioning is strictly executed on a chronological time basis to avoid look-ahead bias. While identical corporate entities naturally appear across the training (1999–2011), validation (2012–2014), and test (2015 onwards) sets to simulate real-world continuous monitoring, entity-level data leakage is strictly precluded. Because the splits are rigidly separated by time, the model is never exposed to a firm's future financial ratios during training, ensuring predictions are made solely on unseen forward-looking data. Financial accounting data possess inherently sequential and complex temporal characteristics. Let vector X denote D distinct financial ratios for firm i . The vector corresponds to the specific reporting year denoted t . A sliding window approach

constructs the historical temporal input sequences. The sequence length L is defined as three historical years. This window captures the trajectory leading to the prediction horizon. The input tensor for a single sample is defined below.

$$X_{i,t} = [X_{i,t-L+1}, X_{i,t-L+2}, \dots, X_{i,t}]. \quad (2)$$

Each element represents the feature vector at a specific time stamp. The resulting tensor shape spans dimensions of length L by D . The corresponding target label indicates status at the future horizon. Data ending in year t predict status in year $t + h$.

3.1.4 | Feature Transformation

The distributions of financial ratios often have nonstationary and heavy tailed features. The extreme statistical outliers tend to distort the standard normalization methods. A nonlinear quantile transformer is used to transform features to a uniform Gaussian distribution. The approach approximates the cumulative distribution function of every feature. The values are then plotted to the inverse-normal distribution quantile function. The mathematical expression of the transformation is given formally below.

$$x_{\text{scaled}} = G^{-1}(F(x)). \quad (3)$$

The F represents the cumulative distribution of feature x . The opposite of G is the quantile function of the normal distribution. Transformed inputs conform to a standard normal distribution having a zero mean. This gradient descent stabilization is used to train deep neural networks.

3.2 | Proposed Model: GST-Net

The GST-Net represents a highly optimized, domain-specific architectural synthesis rather than a fundamentally novel

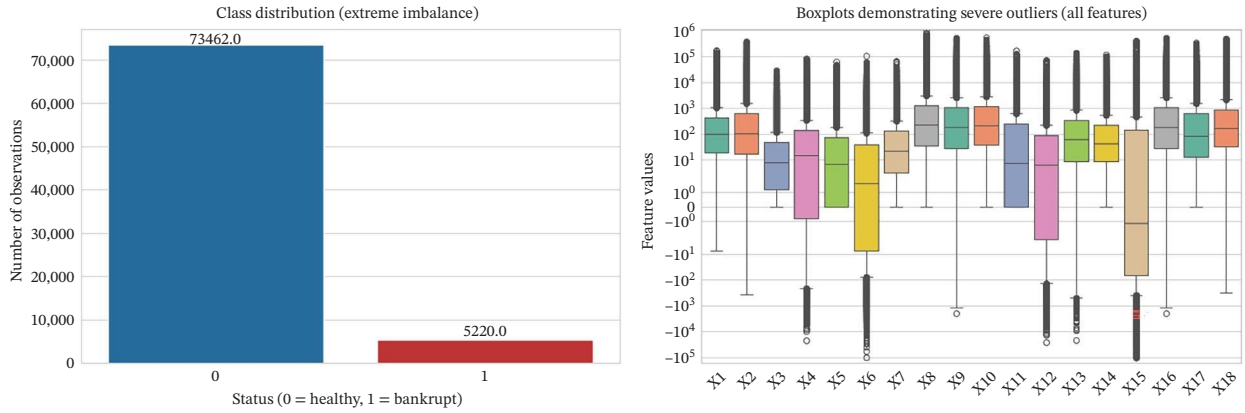


FIGURE 2 | Dataset characterization shows extreme imbalance with 93.37% healthy vs. 6.63% bankrupt and widespread outliers supporting nonlinear quantile transformation.

atomic neural paradigm. The design deals with the financial noise, local volatility, and global trends by synergistically repurposing established spatial-temporal algorithms for the stringent demands of corporate solvency auditing. The design deals with the financial noise, local volatility, and global trends. The architecture has four different and specialized processing modules of computation. The relevant noise of the input signals is suppressed dynamically with a feature gating mechanism. A temporal convolutional branch takes local spectral features of sequences. A transformer encoder branch captures complicated global time dependency structures. Final prediction representations have been incorporated through a fusion and classification module. It is a two-branched approach that is efficient in utilizing complicated financial time series data.

3.2.1 | Feature Gating Mechanism

Financial data are usually full of variables that are redundant; hence, they lower the performance of the models. A gating mechanism that can be learned is added to scale feature weight. The design is inspired by the gated linear unit neural units, indicating that matrix X is the input sequence of financial dimensions. Gating operation calculates weights based on a sigmoid activation function. The mathematical definition of the gating operation is as follows:

$$G(X) = \sigma(XW_g + b_g), \quad (4)$$

where W and b are learnable parameters. The gate values are multiplied with the original input. The resulting feature representation of the gated feature which results is computed through element-wise multiplication.

$$X_{\text{gated}} = X \odot G(X). \quad (5)$$

This operation suppresses irrelevant signals close to zero value effectively. Critical predictive signals are amplified before entering deep network layers.

3.2.2 | Spectral-Temporal Processing (Dual Branch)

The gated features are carried on to a pair of parallel neural processing lines. The former branch makes use of convolutional layers to extract local patterns. One dimensional filter works well

in regard to the time series. The convolution functions use a kernel size of 3 with certain padding conditions. This is carried out to extract spectral features of short-term financial volatility. The second branch makes use of transformer encoders of global temporal context. The positional encodings and linear projection of input vectors are added. The multihead attention mechanism connects various time steps in a direct relationship. Layer normalization and residual connections stabilize deep network training. This is a module that determines the long-term structural tendencies in corporate paths.

3.2.2.1 | Branch 1-Temporal Convolutional Block.

The convolutional branch captures high-frequency variations within data sequences. A 1-dimensional convolutional layer operates along the time dimension. The transformation applies rectification and batch normalization to the output. The mathematical formulation for the convolutional operation is defined below.

$$Z_{\text{cnn}} = \text{ReLU}(\text{BatchNorm}(\text{Conv1d}(X_{\text{gated}}))). \quad (6)$$

The kernel size of three preserves the original temporal sequence length. This architecture extracts spectral features representing rapid financial variable changes. Local patterns like sudden liquidity shocks are effectively detected here.

3.2.2.2 | Branch 2-Transformer Encoder. The transformer branch focuses on modeling global dependency structures effectively. Multihead self-attention mechanisms relate different time steps directly. Input features are projected linearly to the fixed model dimension. Learnable positional encodings are added to the projected input sequence. The mathematical formulation for the embedding step is defined below.

$$E = \text{Linear}(X_{\text{gated}}) + P. \quad (7)$$

The sequence is processed by a transformer encoder layer block. Layer normalization is applied after the residual attention connection sum. The attention mechanism output is formally calculated using this equation.

$$Z' = \text{LayerNorm}(E + \text{MHSA}(E)). \quad (8)$$

A feedforward network processes the intermediate representation using residual connections. This operation adds nonlinearity into the global feature extraction process.

$$Z_{\text{trans}} = \text{LayerNorm}(Z' + \text{FFN}(Z')). \quad (9)$$

In the global context, temporary abnormalities are differentiated from fundamental structural financial decline. The branch does well in detecting trends of long-term deterioration of reliability.

3.2.3 | Feature Fusion and Classification

The independent branch outputs are concatenated along the feature dimension. This operation creates a comprehensive spectral and temporal representation tensor. The mathematical definition of the concatenated tensor is presented below.

$$Z_{\text{concat}} = [Z_{\text{cnn}} || Z_{\text{trans}}]. \quad (10)$$

A linear fusion layer with normalization processes the combined features. Max pooling aggregates the temporal sequence into a fixed vector. This summarization step extracts the most salient predictive signal features.

$$v = \max_t (\text{LayerNorm}(\text{Linear}(Z_{\text{concat}}))). \quad (11)$$

The resulting vector passes through a classifier multilayer perceptron (MLP) block. Dropout and batch normalization are applied to prevent model overfitting. The final sigmoid activation computes the probability of corporate bankruptcy.

$$\hat{y} = \sigma(\text{MLP}(v)). \quad (12)$$

This architecture ensures robust integration of local and global insights. The final output represents the predicted insolvency risk for firms.

3.3 | Optimization and Training Strategy

3.3.1 | Asymmetric Loss Function

Standard binary cross-entropy performs poorly under extreme class imbalance. Models can achieve high accuracy by always predicting the majority class. A weighted cross-entropy loss provides only a partial solution. It treats all negative examples with equal importance during training. Many negative examples are trivial and do not aid learning. The proposed framework implements an asymmetric loss function. This variant of focal loss decouples gradient contributions. Positive and negative samples receive distinct treatment within the loss. The overall loss combines separate positive and negative components.

The mathematical formulation for the asymmetric loss is as follows:

$$L = -yL_+ - (1 - y)L_-, \quad (13)$$

where y represents the true binary label for a given sample. The term L_+ corresponds specifically to positive samples. The term L_- corresponds specifically to negative samples. The positive component L_+ is defined by the equation below.

$$L_+ = (1 - p)(1 - p)^{\gamma_+} \log(p). \quad (14)$$

The variable p denotes the model's predicted probability for bankruptcy. The focusing parameter γ_+ modulates the loss for positive examples. A lower value for γ_+ maintains a stronger gradient signal. This ensures that the model remains attentive to rare positive cases. The negative component L_- employs a modified probability term.

$$L_- = (p_m)^{\gamma_-} \log(1 - p_m). \quad (15)$$

The altered probability p_m uses a hard threshold margin. It is computed as the greater of p -margin and zero. The margin is a good way of screening off easy negative examples. When p is lower than the margin, its loss contribution is zero. The focusing parameter γ_- is considerably greater than γ_+ . This structure is hostile to down-weight remaining easy negatives. In this case, γ_- was determined to be four, and γ_+ was determined to be one. This compels this model to concentrate on hard negatives and infrequent positives. An approximate value of clipping helps avoid numerical instability in the course of training. The asymmetric loss directly solves the severe data imbalance. It puts the emphasis of the model in the critically informative samples. This results in a significant increase in recall and F1 score.

3.3.2 | Ensemble Learning

Ensemble learning methods also help to increase the model robustness and generalization. Each predictive horizon has several autonomous models, which are instantiated. This type of methodology reduces the error, which can be traced to a single model prediction. The model is comprised of 3 different GST-Net models at each horizon; each being trained by a different random initialization. The training plan involves a specialized data-sampling plan that is meant to impose balance in classes.

Mini-batch creation is controlled using a WeightedRandomSampler, similar to the process used during training, which enables mini-batches to perform oversampling of the minority class. This ensures that each model is exposed to a more balanced gradient representation. Optimization is performed using the AdamW optimizer with weight decay. To reduce the risk of overfitting, a cosine annealing schedule adjusts the learning rate over 40 epochs, while early stopping based on the validation set further supports generalization.

The final forecast is a summary of the constituents within an ensemble. The process of aggregations consists of computing a simple arithmetic mean of their probability states, a noise-canceling predictive mean operation that improves the calibration. The formula of the probability of the final ensemble mathematically is as follows:

$$P_{\text{final}} = \frac{1}{N} \sum_{i=1}^N P_i(X), \quad (16)$$

where N equals three for the total number of ensemble members. The term $P_i(X)$ denotes the probability from the i -th model. Input X represents the preprocessed financial sequence data. These probabilistic averages form the final risk score for a given firm.

Aggregation of the predictions is under a final dynamic decision limit. This cutoff level has been adjusted by a systematic grid search on the validation cohort. The optimization process maximizes the macro F1-score within a given parameter space. The calculated optimal threshold is then applicable to the test

data that is independent of the test set. The presence of such an ensemble paradigm ensures a stable and strong performance on unobserved data.

3.4 | Optimization and Training Strategy

There are certain drawbacks in the deployment of conventional deep learning baselines in the context of the prediction of corporate bankruptcy. MLPs themselves distort time sequences, thus losing important time-dependent data. RNNs are highly susceptible to overfitting on short financial time series with a small quantity of time series. Convolutional structures can identify local anomalies, but in a systematic manner, they are not able to understand overall strategic trends. The use of standard transformer models will have inductive biases that are highly sensitive to tabular data and are prone to overfitting to noise. The architecture presented in this paper incorporates the use of convolutional feature extraction and transformer encoder modules. An active gating cell regulates noise and spatial signals selectively based on the salient signals. Parallel processing effectively models both long range and local volatility. This kind of architecture will reduce bounded receptive field effects and make it more resistant to data instability. This fact was empirically validated and demonstrates that the method achieves better predictive consistency than traditional baselines.

In Figure 3, the training dynamics are shown upon 40 different epochs. The gradual decrease in the asymmetric loss is indicated in the upper panel as the optimization attempts are made. The low-level panel shows the simultaneous validation accuracy improvement. These data processing points highlight the effectiveness of the training program to minimize the error in the absence of overfitting.

3.4.1 | MLP

The MLP serves as a fundamental static baseline model. It processes financial time series data as a flat feature vector. The temporal sequence $(X) \in \mathbb{R}^{L \cdot D}$ is transformed as follows:

$$x_{\text{flat}} = \text{Flatten}(X) \in \mathbb{R}^{L \cdot D}. \quad (17)$$

This is a flattened vector form of representation which removes the time-based succession between periods of fiscal years. Thus, the predictive model does not analyze its financial ratios together in any given year and therefore does not have an inherent ability to distinguish the time-based evolution of a series. Consequently, an increasing trend in the level of debt can be indistinguishable to chronic high debt. The architecture that has been widely used, MLP, is several fully connected (L) layers; each layer has a linear mapping followed by a nonlinear activation function on the map.

$$h_1 = \text{ReLU}(W_1 x_{\text{flat}} + b_1). \quad (18)$$

Later layers will always constitute a set of linear and nonlinear transformations. The layer of a terminal uses a sigmoid activation function to generate probabilistic outputs. It is a static approach which overlooks the inherent sequential nature of financial information. Thus, predictability of the model concerning bankruptcy is limited. It cannot capture the dynamic changing that leads to corporate failure. The MLP can be considered an excellent reference point to more sophisticated temporal

modeling methods, and its results reflect the need to have sequential dependencies expressed in an explicit manner.

3.4.2 | RNNs

RNNs are sequential as they process sequential data, step-by-step. They have a secret state record which contains historic data. This state is firstly updated at every new time input. A simple recurrent transition is used in the conventional RNN architecture.

$$h_t = \sigma(W_h h_{t-1} + W_x x_t + b), \quad (19)$$

where σ is often a hyperbolic tangent activation function. The latent state, h_t , is a compressed memory representation. Theoretically, it summarizes all the bits of information of the preceding time steps. In practice, simple RNNs have a problem of vanishing gradient. As a result, it makes the long-term dependencies extremely challenging to capture.

Long short-term memory (LSTM) networks come up to this basic limit by providing a gated architecture which introduces a cell state. The LSTM uses separate input, forget, and output gates, thus allowing to control the memory dynamics precisely. This network, in particular, makes an explicit control over the retention and the release of the information in its hidden state through such mechanisms.

$$\begin{aligned} f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \\ i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \\ \tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C). \end{aligned} \quad (20)$$

The cell state update combines previous memory with new information.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t. \quad (21)$$

The output gate then produces the next hidden state.

$$\begin{aligned} o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \\ h_t &= o_t * \tanh(C_t). \end{aligned} \quad (22)$$

Gated recurrent units offer a slightly simpler gated alternative. They combine the forget and input gates into a single update gate. The GRU also merges the cell state and hidden state. This reduces computational complexity while often performing comparably. Both LSTM and GRU models inherently process sequences sequentially. This sequential nature prevents efficient parallelization during training. Their complex gating mechanisms can overfit on shorter financial sequences. The models also struggle with inverted or very long-term patterns.

3.4.3 | 1D Convolutional Neural Network

The one-dimensional convolutional neural network operates along the temporal dimension. It applies learnable filters across successive time steps within the sequence. Each filter slides across the input to detect localized temporal patterns. This operation for a single filter and time step is defined below.

$$z_t = \sum_{k=0}^{K-1} w_k \cdot x_{t+k} + b, \quad (23)$$

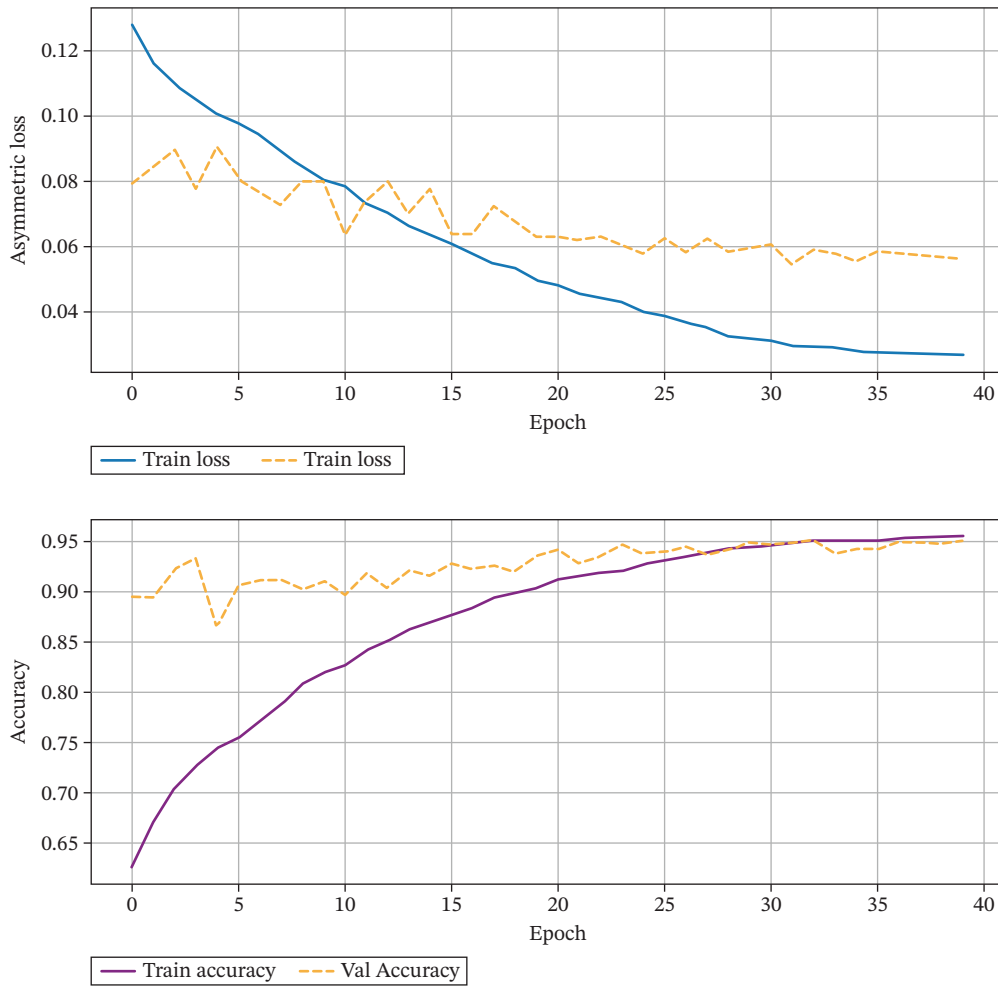


FIGURE 3 | Training dynamics showing the convergence of the asymmetric loss function and the corresponding increase in validation accuracy over forty training epochs.

where w_k represents the filter weights across kernel size k . The term x_{t+k} is the input feature at a specific offset. The bias term b is added to the weighted sum. Multiple filters are applied in parallel to extract diverse features. The resultant feature map contains a variety of local signal descriptions. It is usually followed by the convolutional operation with a nonlinear activation function.

$$A = \text{ReLU}(Z). \quad (24)$$

The pooling layers are also commonly used to reduce dimensionality and encourage the translational invariance. Max pooling, as an example, chooses the best salient activation over a localized time period. The field of view of any particular convolutional layer is inherently limited; therefore, the multiple layers may be stacked up such that the effective field is gradually increased, thus covering a larger amount of contextual information. However, to obtain very long-range dependencies, a large number of layers are required, which can compromise the stability of training and decrease the efficiency of its parameters.

The given convolutional neural network model can be described as having a strong capability of tracking localized anomalies in the framework of financial time series analysis. A sudden reduction in the cash flow produces a strong high-frequency response that the convolutional filters can detect in the small

receptive fields of the filters. On the other hand, the model is troubled when trying to draw a causal connection between a strategic error at the outset and system collapse later; it cannot explicitly correlate observations at different points in time, especially across time frames that extend over several years. This means that there exist wider contextual affiliations and gradually changing trends of multiyears that the network consistently faces difficulties with. The natural result of the design principles of convolutional operations is this inherent constraint, where the space locality principle takes precedence over the time globalization principle.

3.4.4 | Standard Transformer

The traditional transformer design only assumes self-attention mechanisms. It takes into consideration all time locations in an order expansively, hence facilitating effective parallel processing. This parallelizable design therefore gives it a high computational benefit in training. The basic operation on which the architecture is based is the scaled dot-product attention operation.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (25)$$

where Q , K , and V are the query, key, and value matrices, respectively. The derivation of these matrices is in the form of linear projections used to the input sequence. The scaling factor $\sqrt{d_k}$ is used to stabilize the value of the gradients used during the training. There are several such attention mechanisms working in parallel in the multihead attention mechanism. The respective heads then get to know how to cater to different aspects of the time relationships that are in the data.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O. \quad (26)$$

The calculation of each attention head is performed independently in this architecture. The output is then joined, and a linear projection is applied on them. Such a construction provides an internationally desirable field that covers the length of the sequence. This also means that, at any level of time, each position can have a direct effect on the representation of any other position in the sequence. The model is shown to have higher effectiveness in the capture of long-range dependencies and complex structures.

However, the natural designs of transformers do not have inductive biases that favor tabular data. More specifically, they lack a predetermined affinity to local interfeature interactions, despite the fact that the financial volatility presents itself as temporally limited patterns. This generally involves a large amount of data to discover such local structures, and with comparatively small financial datasets, pure transformers can just overfit to noise. In addition, the model does not explicitly focus on priority or kill off incorrectly trading ratios in the feature space by focusing on the homogenous nature of the feature space implicitly. The end result of this homogenous treatment may be poor convergence dynamics and low performance in generalization.

3.5 | Experimental Methodology and Evaluation

Correctness is a deceptive measure when there is a strong imbalance in the classes. The evaluation framework would, therefore, focus more on macro F1 score and the Matthews correlation coefficient (MCC). The AUC-ROC is a measure of discriminative ability. The kappa statistic given by Cohen is used to explain the possible possibility of agreement in binary classification by chance. The calibration of decision thresholds is dynamic to make the best that they can on the validation set. The purpose of such calibration is to increase practical usefulness in the area of financial decision-making. An ensemble approach reduces the variance and provides predictive strength. Each individual prediction horizon is trained separately on three different models. A weighted random sampler helps to achieve balance of classes in the training batches. The AdamW optimizer uses cosine annealing to schedule the learning rate. The average probability estimates of all the ensembles characterize the final predictions.

To ensure a strictly fair comparative analysis, all baseline models were trained using an identical preprocessing pipeline with global standard scaling and identical chronological data splits. Hyperparameter optimization was performed through Bayesian optimization using the Optuna framework with 20 optimization trials for each architecture. The search space included learning rates ranging from 1×10^{-4} to 5×10^{-3} on a logarithmic scale, hidden dimensions of 32, 64, and 128, and dropout rates between

0.1 and 0.5. For tree-based models such as XGBoost, the maximum tree depth was optimized within the range of 3–7, while the number of estimators was searched between 50 and 150. Early stopping with a patience of 5 epochs based on validation loss and validation AUC was consistently applied to all deep learning baselines to reduce overfitting while ensuring adequate model convergence. The optimal hyperparameter configurations for all machine learning and deep learning models across different prediction horizons are presented in Table 2.

To guarantee full reproducibility of the GST-Net framework, all granular implementation parameters, architectural dimensions, and optimization schedules are explicitly formalized in Tables 3 and 4. The model is optimized utilizing AdamW with a base learning rate of $5e^{-4}$, regulated by a cosine annealing schedule over 40 epochs. To actively combat the severe class imbalance inherent to bankruptcy data, the training pipeline utilizes a WeightedRandomSampler in conjunction with a rigorously calibrated asymmetric loss function, prioritizing hard negative mining with a negative gamma of 4. Furthermore, the dual-branch structural dimensions are fixed at an embedding size of 128 to balance computational efficiency with the capacity required to model highly complex, long-horizon financial representations.

3.5.1 | Metrics

To reduce acute imbalance in classes, model evaluation has a multiplicity of measures. Traditional accuracy provides inaccurate and overly positive evaluation of performance. The main assessment scale is the macro F1 score that gives a moderated judgment. This measure is a combination of the harmonic mean of the precision and the recall on single classes. The final score averages these values across bankruptcy and healthy classes.

$$F1_{\text{class}} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (27)$$

$$\text{MacroF1} = \frac{F1_{\text{positive}} + F1_{\text{negative}}}{2}.$$

This metric assigns equal importance to performance on both minority and majority classes. The ROC-AUC is also reported. The ROC-AUC evaluates the model's discriminative ability across all classification thresholds. It represents the probability that a randomly selected positive instance is ranked higher than a randomly selected negative instance.

$$\text{ROC} - \text{AUC} = \int_0^1 \text{TPR}(\text{FPR})d\text{FPR}. \quad (28)$$

MCC provides a complete scalar value. The MCC does include true positives, false positives, true negatives, and false negatives in its formulation. When the model turns out to be unique to all classes, a high score is obtained.

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (29)$$

Cohen's kappa statistic further adjusts for chance classifier agreement. The method is especially beneficial when it is used with the datasets that have a strong imbalance in the classes. The metrics are calculated solely on a held-up temporal test set, thus

TABLE 2 | Optimal hyperparameter configurations for machine learning and deep learning models across different prediction horizons.

Horizon	Model	Best parameters
1-Year	XGBoost	{‘learning_rate’: 0.284101534994126, ‘max_depth’: 6, ‘n_estimators’: 124}
	TabNet	{‘n_da’: 32, ‘lr’: 0.01895889473372275}
	TabPFN	{‘lr’: 0.0017065039532206232, ‘batch_size’: 256}
	MLP	{‘lr’: 0.0012178930552730922, ‘hidden_dim’: 128, ‘dropout’: 0.374208379104926}
	RNN	{‘lr’: 0.0024555888040567408, ‘hidden_dim’: 64}
	LSTM	{‘lr’: 0.002208146552185645, ‘hidden_dim’: 64}
	GRU	{‘lr’: 0.0016178213005275707, ‘hidden_dim’: 64}
	1D-CNN	{‘lr’: 0.0012095831180803072, ‘hidden_dim’: 32}
	Transformer	{‘lr’: 0.00038762321447639126, ‘hidden_dim’: 128}
2-Year	XGBoost	{‘learning_rate’: 0.29489097723956237, ‘max_depth’: 7, ‘n_estimators’: 88}
	TabNet	{‘n_da’: 24, ‘lr’: 0.058615675924496305}
	TabPFN	{‘lr’: 0.0006918672791245567, ‘batch_size’: 128}
	MLP	{‘lr’: 0.001225454529848914, ‘hidden_dim’: 128, ‘dropout’: 0.10826639205406713}
	RNN	{‘lr’: 0.0010984188616888647, ‘hidden_dim’: 128}
	LSTM	{‘lr’: 0.0018817513461972832, ‘hidden_dim’: 64}
	GRU	{‘lr’: 0.0028488279480132334, ‘hidden_dim’: 64}
	1D-CNN	{‘lr’: 0.003523911047879568, ‘hidden_dim’: 128}
	Transformer	{‘lr’: 0.00019746414675240675, ‘hidden_dim’: 128}
3-Year	XGBoost	{‘learning_rate’: 0.11416596940762931, ‘max_depth’: 7, ‘n_estimators’: 120}
	TabNet	{‘n_da’: 32, ‘lr’: 0.06984915958458938}
	TabPFN	{‘lr’: 0.002374717354009401, ‘batch_size’: 128}
	MLP	{‘lr’: 0.002814738407434614, ‘hidden_dim’: 64, ‘dropout’: 0.12990587242860344}
	RNN	{‘lr’: 0.00364243313399458, ‘hidden_dim’: 32}
	LSTM	{‘lr’: 0.002023817830369197, ‘hidden_dim’: 64}
	GRU	{‘lr’: 0.00355886539602373, ‘hidden_dim’: 32}
	1D-CNN	{‘lr’: 0.0032399335677340047, ‘hidden_dim’: 128}
	Transformer	{‘lr’: 0.00044968494887014164, ‘hidden_dim’: 128}

protecting the integrity of the testing. Therefore, this methodology ensures that an effective and realistic model generalization appraisal is made.

3.5.2 | Threshold Optimization

The default probability threshold that is used in conventional classification practice is 0.5. This threshold is often not

optimal where one is using highly biased financial data. Binary decision-making requires the model to be calibrated with the resulting probabilities that are generated by the model. The choice of a threshold will have a direct effect on the trade-off between precision and recall. False negatives and false positives may have costs put on them by financial decision-makers. The model is optimized dynamically to the operational setting.

TABLE 3 | Training, optimization, and hyperparameters.

Category	Parameter	Value
Optimization	Optimizer	AdamW
	Learning rate (α)	5×10^{-4}
	Weight decay	1×10^{-5}
	Learning schedule	Cosine Annealing, $T_{\max} = 40$ epochs
	Batch size	128
	Epochs	40
Regularization	Dropout (general)	0.2 (applied in Transformer and CNN blocks)
	Dropout (classifier)	0.3 (applied in final dense classification layer)
Loss function	Method	Asymmetric loss
	γ_{neg} (negative gamma)	4 (focusing on easy negative samples)
	γ_{pos} (positive gamma)	1 (focusing on positive samples)
	Clipping (ϵ)	0.05 (prevents numerical instability)
Inference	Ensemble count	3 (number of models averaged for final prediction)
	Decision threshold	Dynamic, 0.2–0.85 (optimized via grid search on validation F1)

TABLE 4 | Data preprocessing and model architecture specifications.

Category	Parameter	Value/setting
Data setup	Data scaling	QuantileTransformer → normal distribution
	Class balancing	WeightedRandomSampler
CNN branch	Convolution type	1D temporal conv
	Kernel size/padding	3/1
	Feature dimension	128
Transformer branch	Embedding dimension	128
	Attention heads	4
	Encoder layers	3
	FFN dimension	256
Fusion and output	Fusion strategy	Concatenation + linear
	Classification head	256 → 128 → 1 (FC layers with BatchNorm + ReLU)
	Activation function	Sigmoid (output probability)

The optimal choice level is established via a grid search. The practical space of search is within the 0.2 and 0.8 range. All candidate thresholds change probabilities into binary form. Each threshold is evaluated on an independent set of validation. The macro F1 score is the object of optimization.

$$\tau^* = \operatorname{argmax}_{\tau \in [0.2, 0.8]} \text{F1}_{\text{Macro}}(\text{Val}, \tau). \quad (30)$$

In this work, the parameter is defined as τ . The relevant calculation is a macro F1 score that is calculated at this threshold. To choose the maximum score, an argmax operation is used to choose the threshold with the maximum score. The threshold level optimized in each of the trained models is given as τ^* . It does not use it retrospectively on the training data.

This optimal threshold is used to evaluate the final performance on the test set. The test set probabilities are binarized based on a threshold τ , rather than the default value of 0.5. This process improves the model's practical applicability for stakeholders. It ensures that the reported performance more closely reflects a realistic operational setting. In imbalanced classification tasks, threshold optimization is a critical step for improving decision quality.

4 | Comparative Performance Analysis

4.1 | Model Performance Overview Across Prediction Horizons

The proposed GST-Net exhibits better performance in all the horizons considered. Table 5 is a full report on performance metrics. The prediction accuracy of one-year is 95.03. Given the severe structural class imbalance in the corporate insolvency dataset, where 93.37% of firms are financially healthy and only 6.63% are bankrupt, conventional accuracy provides a misleading assessment of predictive performance. Therefore, the evaluation framework primarily emphasizes imbalance sensitive metrics, including the MCC, and the macro F1 score. As reported in Table 6, the proposed GST-Net achieves an MCC of 0.5704 and a macro F1 score of 0.7838 at the challenging 3-year prediction horizon. These highly discriminative metrics demonstrate that the superior performance of the proposed framework is not driven by majority class dominance but instead reflects its strong ability to

preserve both precision and recall for the minority bankruptcy class across long-term forecasting horizons.

The 2-year horizon shows further improvements on accuracy and precision. The accuracy increases to 95.54 in the same period. The accuracy becomes much higher at 67.30. Recall, however, decreases slightly only to 55.70. The F1 score is a little higher in a macroaverage of 79.29. The maximum value of ROC-AUC is 93.54. The MCC also reaches its highest coefficient at 58.90, which means that there is a balanced classification performance.

Exceptional classification accuracy is maintained with the 3-year horizon. The long-term prediction has a value of accuracy of 95.95. Precision is 64.77 while recall is 54.01. The macro F1 score stands at 78.38, which implies the steady performance. The ROC-AUC remains robust at 92.39. The MCC is 57.04, which supports the trustworthy 3-year projections.

To further validate the academic contribution of the proposed framework, GST-Net was systematically benchmarked against recent deep learning and multimodal SOTA models in bankruptcy prediction. For example, the NLP-enhanced transformer architecture by Lombardo et al. [24] achieved a 1-year accuracy of 87.50% by integrating accounting time-series data with SEC textual disclosures. Likewise, the multimodal tabular-text transformer proposed by Bonnier [31] demonstrated competitive uncertainty-aware prediction by combining structured numerical inputs with textual features. In comparison, GST-Net significantly outperforms both approaches across longer prediction horizons. At the demanding 3-year forecasting interval, GST-Net attains an accuracy of 95.95% and an ROC-AUC of 92.39%. These results confirm that the proposed dynamic feature-gating mechanism and dual-branch spectral-temporal architecture are more effective in capturing long-term structural degradation while mitigating short-term financial noise, outperforming existing SOTA methods.

4.2 | Minority Class Effectiveness and Imbalanced Learning Metrics

While aggregate metrics such as ROC-AUC ranging from 0.9239 to 0.9354 and overall accuracy above 95% demonstrate strong general discrimination capability, corporate bankruptcy prediction inherently involves extreme class imbalance. Therefore, we emphasize evaluation metrics that strictly penalize poor

TABLE 5 | Performance of the proposed GST-Net across different prediction horizons.

Horizon	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
1-Year	0.9503	0.6016	0.6016	0.7875	0.9330	0.6458	0.5751
2-Year	0.9554	0.6730	0.5570	0.7929	0.9354	0.6576	0.5890
3-Year	0.9595	0.6477	0.5401	0.7838	0.9239	0.6317	0.5704

minority class performance, particularly PR AUC, the MCC, and class-specific F1 scores. Table 7 provides a detailed classification report for both bankrupt and healthy classes.

Predicting rare corporate failures several years in advance inherently involves significant financial uncertainty. Nevertheless, GST-Net demonstrates strong minority class detection capability. At the 1-year horizon, the model achieves balanced minority precision and recall of 0.6016. Notably, even at the more difficult 3-year horizon, it maintains a minority precision of 0.6477 and an F1 score of 0.5890. In highly imbalanced financial datasets where baseline models often exhibit near zero recall for the minority class or precision below 0.20, these outcomes represent a meaningful predictive advantage. The consistently strong MCC values ranging from 0.5704 to 0.5890 and PR AUC scores further validate that the high accuracy is not driven by majority class bias but instead reflects the effectiveness of the asymmetric loss

function and gating mechanism in identifying rare distress signals.

4.3 | Superiority of Proposed GST-Net Architecture

The suggested model is proven to be superior to any of the baseline deep learning structures. Table 6 shows comparative results in 1-, 2-, and 3-year horizons. The MLP baseline has a medium level of performance deterioration over time. Its accuracy reduces to 54.63%. The macro F1 score dropped to 42.38% out of 56.24%. Such findings highlight the weakness in the MLP to model time-related dependencies.

There is an unstable performance profile that RNNs and their variations have. Basic RNNs vary in accuracy significantly with different prediction horizons, whereas the LSTM models show an imprecise overview with regard to recall and precision. Gated

TABLE 6 | Performance comparison across different prediction horizons.

Horizon	Model	Accuracy	Macro F1	ROC-AUC	MCC	Inference time (s)
1-Year	XGBoost	0.9131	0.6664	0.8284	0.3349	0.0360
	TabNet	0.8633	0.5697	0.7131	0.1522	0.3800
	TabPFN	0.7056	0.5070	0.6914	0.1457	0.1350
	MLP	0.7747	0.5546	0.7607	0.2044	0.1206
	RNN	0.4874	0.4007	0.7416	0.1508	0.1414
	LSTM	0.5284	0.4233	0.7438	0.1514	0.1445
	GRU	0.7473	0.5392	0.7487	0.1942	0.1384
	1D-CNN	0.7698	0.5469	0.7187	0.1869	0.1344
	Transformer	0.7060	0.5242	0.7885	0.2091	0.1804
	GST-Net (proposed)	0.9503	0.7875	0.9330	0.5751	0.4879
2-Year	XGBoost	0.9235	0.6528	0.8206	0.3059	0.0260
	TabNet	0.5003	0.4008	0.7089	0.1181	0.3460
	TabPFN	0.5129	0.4106	0.7082	0.1375	0.1390
	MLP	0.7928	0.5657	0.7680	0.2208	0.1199
	RNN	0.4878	0.3994	0.7517	0.1541	0.1282
	LSTM	0.4524	0.3772	0.7494	0.1407	0.1293
	GRU	0.5427	0.4289	0.7485	0.1526	0.1276
	1D-CNN	0.5295	0.4229	0.7501	0.1582	0.1191
	Transformer	0.6012	0.4644	0.7925	0.1816	0.1516
	GST-Net (proposed)	0.9554	0.7929	0.9354	0.5890	0.4219
3-Year	XGBoost	0.9128	0.6612	0.8446	0.3294	0.0290
	TabNet	0.3312	0.2937	0.6582	0.0916	0.2950
	TabPFN	0.3903	0.3317	0.6714	0.0905	0.1220
	MLP	0.7789	0.5372	0.7308	0.1614	0.0993
	RNN	0.8021	0.5471	0.7340	0.1622	0.1144
	LSTM	0.4691	0.3830	0.7388	0.1321	0.1314
	GRU	0.7051	0.5079	0.7290	0.1705	0.1288
	1D-CNN	0.8271	0.5595	0.7308	0.1686	0.1075
	Transformer	0.7864	0.5611	0.7951	0.2265	0.1453
	GST-Net (proposed)	0.9595	0.7838	0.9239	0.5704	0.3735

Note: Bold values indicate results of the proposed model.

TABLE 7 | Detailed classification report across prediction horizons.

Horizon	Class	Precision	Recall	F1 score
1-Year	Bankrupt	0.6016	0.6016	0.6016
	Healthy	0.9735	0.9735	0.9735
	Accuracy			0.9503
	Macro avg	0.7875	0.7875	0.7875
	Weighted avg	0.9503	0.9503	0.9503
2-Year	Bankrupt	0.6730	0.5570	0.6095
	Healthy	0.9708	0.9819	0.9763
	Accuracy			0.9554
	Macro avg	0.8219	0.7695	0.7929
	Weighted avg	0.9521	0.9554	0.9534
3-Year	Bankrupt	0.6477	0.5401	0.5890
	Healthy	0.9741	0.9833	0.9787
	Accuracy			0.9595
	Macro avg	0.8109	0.7617	0.7838
	Weighted avg	0.9565	0.9595	0.9577

recurrent units fail to maintain competitive performance beyond 1 year. The series of models used below have low efficacy when faced with the abbreviated financial series used in the present investigation. The gating mechanisms, which are sophisticated in the nature and appearance of these architectures, appear to promote overfitting within the limited time frame.

Performance visualizations in the form of confusion matrix and receiver operating characteristic curves are presented in Figure 4. The matrix on the left outlines classification results for the 3-year prediction horizon. The curves on the right are a contrast in discriminatory power between 1-, 2- and 3-year forecasts. High values for AUC indicate high generalization on all timeframes.

4.4 | Performance of Advanced Tabular Baselines

To ensure a rigorous evaluation against contemporary SOTA tabular models, the comparative framework was expanded to include XGBoost, TabNet, TabPFN, and an attention-enhanced CNN. When evaluated through the lens of strict, imbalance-aware metrics, the limitations of contemporary SOTA baselines become sharply apparent. As summarized in Table 7, among the optimized tabular models, XGBoost demonstrated the highest resilience, peaking at a 3-year ROC-AUC of 0.8446. Nevertheless, it falls significantly short of the GST-Net in strictly imbalanced metrics, yielding a maximum 3-year MCC of only 0.3294 compared to the GST-Net's 0.5704. Strikingly, advanced deep tabular models, specifically TabNet and TabPFN, exhibited catastrophic performance degradation in minority class recognition. TabNet's macro F1 collapsed to 0.2937 at the 3-year horizon, while TabPFN struggled to surpass an MCC of 0.0905. Furthermore, standard sequential deep learning baselines such as the 1D-CNN and transformer plateaued at MCCs of 0.1686 and 0.2265, respectively. This pronounced underperformance exposes a critical limitation in both strictly tabular and standard sequential architectures when confronted with severe financial data imbalance. The substantial performance improvements exhibited by the GST-Net across these rigorous metrics empirically validate the efficacy of our dual-branch design and asymmetric loss function in isolating rare structural distress signals.

The performance of the fully optimized baselines compared to the proposed GST-Net is summarized below in Table 6. To ensure a rigorous evaluation against contemporary SOTA tabular models, the comparative framework was expanded to include XGBoost, TabNet, and TabPFN, alongside optimized standard deep learning baselines. Among these, XGBoost demonstrated the highest resilience. The statistically significant performance margins of the GST-Net over these optimized baselines empirically substantiate the utility of our dual-branch design, which uniquely bridges localized tabular feature extraction with global sequential dependency tracking.

4.5 | Statistical Significance and Robustness Evaluation

To ensure that the reported performance improvements are empirically rigorous and not attributable to random variance, a one-sided Wilcoxon signed-rank test was applied to the macro F1 scores across 15 combined evaluation folds spanning the 1-to-3-year horizons. The analysis yielded a statistically significant p value of less than 0.05, confirming that GST-Net consistently outperforms the comparison baselines. Furthermore, to explicitly quantify the model's systemic robustness against financial data corruption and input volatility, a Gaussian noise injection test was executed during the inference phase. As illustrated in Table 8, the architecture exhibits remarkable resilience to stochastic perturbation. Even under simulated noise, the model maintains high predictive integrity, with the 3-year forecasting horizon retaining an accuracy of 0.9545 and an ROC-AUC of 0.9049. This sustained performance empirically supports the efficacy of the proposed feature gating mechanism, demonstrating its capacity to dynamically filter unstructured market noise while preserving critical long-term structural distress signals.

4.6 | Ablation Study of Architectural Components

To isolate the predictive value and empirical contribution of each architectural innovation, a comprehensive ablation study was conducted across all three forecasting horizons. Table 9 details the performance degradation observed when systematically removing the dual-branch architecture, the ensemble learning mechanism, the active gating module, and the asymmetric loss function from the baseline GST-Net. The results reveal that no single component acts as the sole driver of the model's predictive superiority; rather, they function in strict synergy. Removing either the asymmetric loss or the ensemble framework severely compromises the model's ability to handle class imbalance, causing the macro F1 score to plummet from approximately 0.79 down to the low 0.60s across all horizons. This highlights their indispensable role in preventing majority-class overfitting. Furthermore, eliminating the dual-branch architecture or the feature gating mechanism drastically reduces the overall ROC-AUC. Notably, the absence of the dual-branch configuration drops the 3-year ROC-AUC from 0.9239 to 0.7786. This performance degradation confirms that the concurrent extraction of local spectral volatility and global temporal dependencies is a critical factor in suppressing short-term financial noise and capturing the long-term structural trends required for robust multiyear solvency forecasting.

To rigorously validate the effectiveness of the GST-Net architecture, we extended our comparative analysis beyond standard

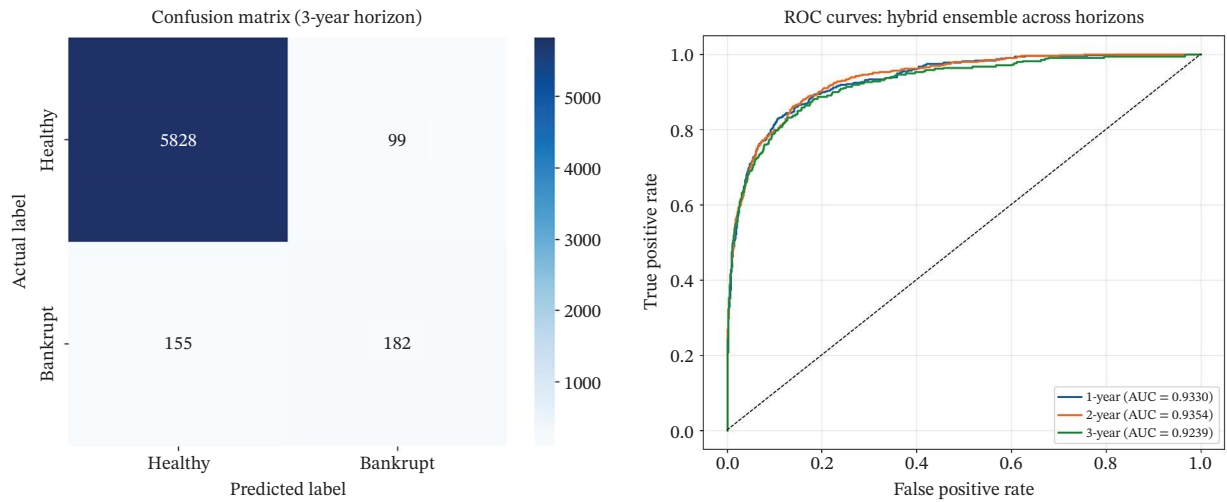


FIGURE 4 | Performance visualizations displaying the confusion matrix for the 3-year prediction horizon and ROC curves comparing model performance across 1-year, 2-year, and 3-year forecasts.

deep learning baselines to include highly optimized, domain-specific models from the recent literature. As shown in Table 10, we benchmarked the 3-year prediction performance of GST-Net against an optimized XGBoost framework, a dynamic graph convolutional network with temporal aggregation (DGCN-TA), an adversarially robust MLP utilizing FGSM, a fuzzy shadowed support vector machine, and a fuzzified neural network. The empirical results demonstrate the distinct advantage of the proposed framework in handling heavily skewed financial sequences. While the optimized XGBoost and fuzzy SVM models achieve exceptionally high overall accuracies of 0.9648 and 0.9620, respectively, these metrics are highly misleading in the context of extreme data imbalance. Their corresponding F1 scores crash to 0.4286 and 0.1764, alongside severe drops in recall, revealing a catastrophic failure to identify the minority bankruptcy class reliably. The DGCN-TA provides robust competition with a recall of 0.7489 and an F1 score of 0.6892, while the MLP with FGSM achieves a balanced but lower overall performance profile. Ultimately, the GST-Net successfully circumvents the accuracy–recall trade-off that cripples the benchmarked models. It achieves an accuracy of 0.9595 while simultaneously recording the highest F1 score of 0.7838 and a strong recall of 0.7617. This confirms that the integration of dual-branch spectral–temporal feature extraction, alongside the dynamic asymmetric loss function, produces a markedly superior and more reliable model for long-horizon risk analysis compared to the current domain-specific SOTA. The references corresponding to these new baselines have also been incorporated into the bibliography.

5 | Explainable AI (XAI) Framework

An explainable artificial intelligence architecture provides assurance to stakeholders in the financial industry through model transparency. The model balances predictive performance with regulatory auditability. It integrates both global and local post hoc explanation techniques. Global techniques identify features that maintain consistent importance across the entire dataset, while local methods explain individual predictions associated with specific firms or time periods. This dual approach addresses the heterogeneous interpretability requirements of different stakeholders. To establish a foundational context for these subsequent interpretability analyses, the precise definitions and corresponding nomenclatures of the fundamental financial indicators scrutinized by the GST-Net model are comprehensively cataloged in Table 11.

5.1 | SHAP for Global Feature Importance Analysis

SHAP values can give a single measure of contribution of the features in predictions. It is based on the theory of cooperative game and Shapley values. It measures the contribution of each feature to the output of the predictive model that is marginal. It is defined as follows: The Shapley value of feature $\{j\}$ is

$$\phi_j(val) = \sum_{S \subseteq \{1, \dots, D\} \setminus \{j\}} \frac{|S|!(D - |S| - 1)!}{D!} (f(S \cup \{j\}) - f(S)), \quad (31)$$

where S represents a subset of all available features excluding j . The term D is the total number of input features in the model.

TABLE 8 | Model robustness under Gaussian noise injection.

Model/setting	Horizon	Accuracy	F1/macro F1	ROC–AUC
Proposed model	1-Year	0.9503	0.7875	0.9330
Robustness (Gaussian noise)		0.9405	0.7305	0.9015
Proposed model	2-Year	0.9554	0.7929	0.9354
Robustness (Gaussian noise)		0.9436	0.7277	0.9110
Proposed model	3-Year	0.9595	0.7838	0.9239
Robustness (Gaussian noise)		0.9545	0.7461	0.9049

TABLE 9 | Ablation study results.

Horizon	Model	Accuracy	Macro F1	ROC-AUC
1-Year	Without dual-branch	0.8969	0.6189	0.7916
	Without ensemble	0.8906	0.6087	0.7861
	Without gating	0.9063	0.6322	0.8049
	Without asymmetric loss	0.9043	0.6312	0.8104
	Proposed GST-Net	0.9503	0.7875	0.9330
2-Year	Without dual-branch	0.9026	0.6099	0.7960
	Without ensemble	0.8971	0.6192	0.7922
	Without gating	0.8936	0.6155	0.8115
	Without asymmetric loss	0.8847	0.6218	0.8221
	Proposed GST-Net	0.9554	0.7929	0.9354
3-Year	Without dual-branch	0.9049	0.6012	0.7786
	Without ensemble	0.9037	0.6007	0.7781
	Without gating	0.9041	0.6220	0.8082
	Without asymmetric loss	0.9168	0.6095	0.8003
	Proposed GST-Net	0.9595	0.7838	0.9239

The function f is the trained GST-Net model's prediction function. The Shapley value ϕ_j is a weighted average of marginal contributions. This calculation guarantees properties of local accuracy and consistency. Kernel SHAP provides a model-agnostic approximation for complex deep networks. It generates a simplified linear explanation model for the original prediction. Beeswarm plots visualize the global distribution of SHAP values. These plots show not just which financial ratios are important overall but also illustrate how a feature's value impacts the bankruptcy risk score. High versus low values for the same feature can have opposite effects.

Figure 5 presents global and local SHAP-based feature attribution analyses mapped to their corresponding financial indicators to yield actionable insights. The summary chart on the right indicates that the GST-Net model places predominant emphasis on structural profitability and market valuation when producing its three-year forecasts; specifically, EBITDA (X4) and market value (X8) emerge as the most influential global predictors. From a corporate finance perspective, this aligns with fundamental solvency theory: EBITDA directly measures core operational profitability and raw cash-generation capacity independent of capital structure, while Market Value encapsulates the collective market foresight regarding the firm's future viability and acts as a critical equity buffer to absorb prolonged distress.

Crucially, extending this SHAP analysis across all prediction horizons reveals a highly logical financial evolution in feature dominance. For the immediate 1-year horizon, short-term

liquidity indicators such as current assets (X1) and total current liabilities (X14) exhibit dominant predictive power, reflecting the reality that acute cash crunches and working capital deficits are the proximate triggers of immediate Chapter 11 filings. However, as the forecasting horizon extends to 2 and 3 years, these transient liquidity metrics are systematically superseded by structural solvency and profitability indicators most notably EBITDA (X4), net income (X6), and total long-term debt (X11). This temporal shift confirms that the GST-Net successfully learns to dynamically differentiate between short-term liquidity shocks and long-term structural decay, prioritizing fundamental operational viability when assessing distant solvency risks. Furthermore, the inclusion of historical lagged features (X4_t0, X4_t-1, and X4_t-2) among the top global contributors demonstrates that the temporal convolutional and transformer branches effectively capture multiyear deterioration trajectories, attenuating single-year anomalous noise to project genuine corporate decline.

5.2 | LIME for Local Case-Specific Explanations

Local interpretable model-agnostic explanations analyze individual firm predictions. LIME approximates the complex GST-Net decision boundary locally. It learns an interpretable linear model around a specific firm's data point. This linear model is faithful to the original classifier's behavior locally. The explanation is obtained by solving the following optimization problem:

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \mathcal{L}(f, g, \pi_x) + \Omega(g). \quad (32)$$

TABLE 10 | Performance comparison with state-of-the-art domain-specific models.

Model	Method	Accuracy	Recall	F1 score
Revalina Suyatno et al. [38]	Optimized XGBoost	0.9648	0.6962	0.4286
Yue et al. [39]	DGCN-TA	0.8789	0.7489	0.6892
Xing et al. [40]	MLP + FGSM	0.7826	0.7826	0.7826
Abdelhamid et al. [41]	Fuzzy + SVM	0.9620	0.1667	0.1764
Sinkovskiy et al. [42]	Fuzzified neural network	0.8210	0.4320	0.4803
Proposed	GST-Net (3-Year)	0.9595	0.7617	0.7838

Note: Bold values indicate results of the proposed model.

TABLE 11 | Core financial variables and definitions utilized in the predictive framework.

Variable name	Description
X1	Current assets
X2	Cost of goods sold
X3	Depreciation and amortization
X4	EBITDA
X5	Inventory
X6	Net income
X7	Total receivables
X8	Market value
X9	Net sales
X10	Total assets
X11	Total long-term debt
X12	EBIT
X13	Gross profit
X14	Total current liabilities
X15	Retained earnings
X16	Total revenue
X17	Total liabilities
X18	Total operating expenses

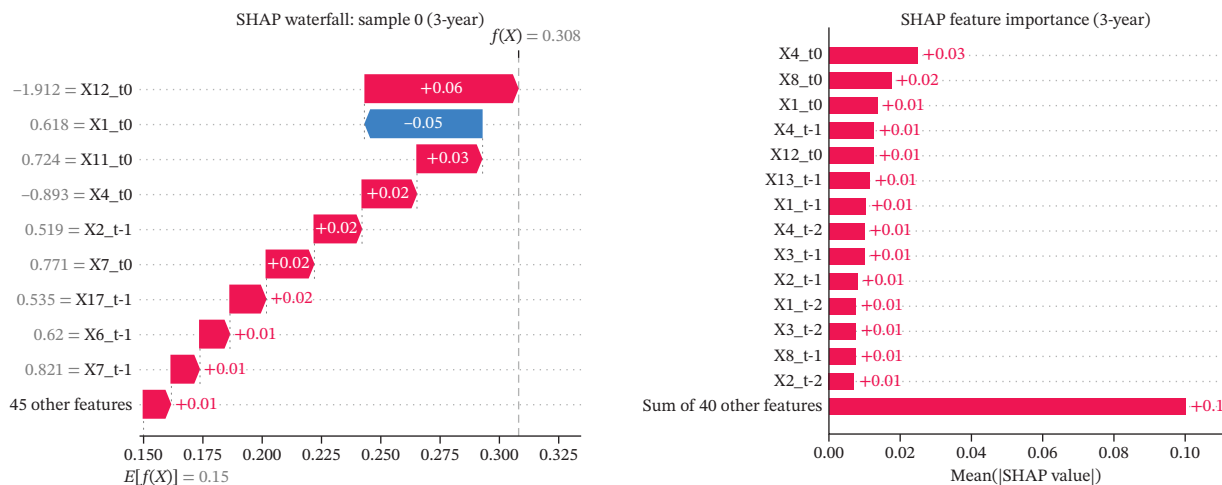
In this formulation, x is the instance being explained for a specific firm. The set G contains all possible interpretable linear models. The loss function $\mathcal{L}$ measures how unfaithful g is to f . The proximity measure π_x defines the locality around the instance x . The term $\Omega(g)$ penalizes the complexity of the explanation g . The method perturbs the original input and observes prediction changes. It weights these new samples by their proximity to the original instance. A sparse linear model is then fit to these weighted perturbed samples. The methodology creates an enumeration of contributory variables relating to the entity in question in a human-readable format. For example, one of the report's main risk determinants might be deficient retained earnings. Such meticulous insight makes financially informed decision and audit process easier.

The analysis presented in Figure 6 employs LIME to interrogate localized decision boundaries for prototypical high-risk and low-risk profiles, thereby furnishing the precise audit trails required by financial regulators. In the high bankruptcy risk firm depicted in the left panel, the explainer isolates pronounced vulnerabilities in capital structure and valuation. Deficiencies in market value X8_t0 together with historical anomalies in total current liabilities X14_t-2 and total operating expenses X18_t-2 provide robust support for the insolvency prediction as reflected by the positive contribution bars. The model concurrently evaluates these adverse signals against historical cost of goods Sold X2 and total long-term debt X11_t-2 trajectories.

By contrast, the low bankruptcy risk firm shown in the right panel exhibits an entirely different attributional profile. The LIME explanation emphasizes the stabilizing influence of sustained debt management and accumulated profitability. A resilient historical pattern of retained earnings X15_t-1 and disciplined total long-term debt across t0 t-1 and t-2 operate as dominant stabilizing factors that outweigh a localized decline in recent EBIT X12_t0. By translating abstract model weights into discrete accounting line items, GST-Net effectively functions as an automated forensic auditor, ensuring that when a firm is flagged for potential insolvency at 3-year horizon stakeholders are immediately presented with the exact combination of operational liquidity and leverage metrics driving the alert.

6 | Limitations and Future Work

The current analysis is solely dependent on the quantitative inputs of financial accounting ratios and thus does not yet capitalize on qualitative textual disclosures extracted from annual reports. Given that the broader literature has already proven the superior performance of multimodal frameworks utilizing financial dictionary-based sentiment, a primary limitation of the current GST-Net is its purely numerical nature. Future research will focus on extending the GST-Net architecture into a fully multimodal framework. By building upon these established methodologies for extracting textual sentiment, we intend to synergize our long-horizon spectral-temporal representations

**FIGURE 5** | Global and local interpretability analysis using SHAP, featuring a waterfall plot for individual sample attribution and a summary plot for global feature importance.

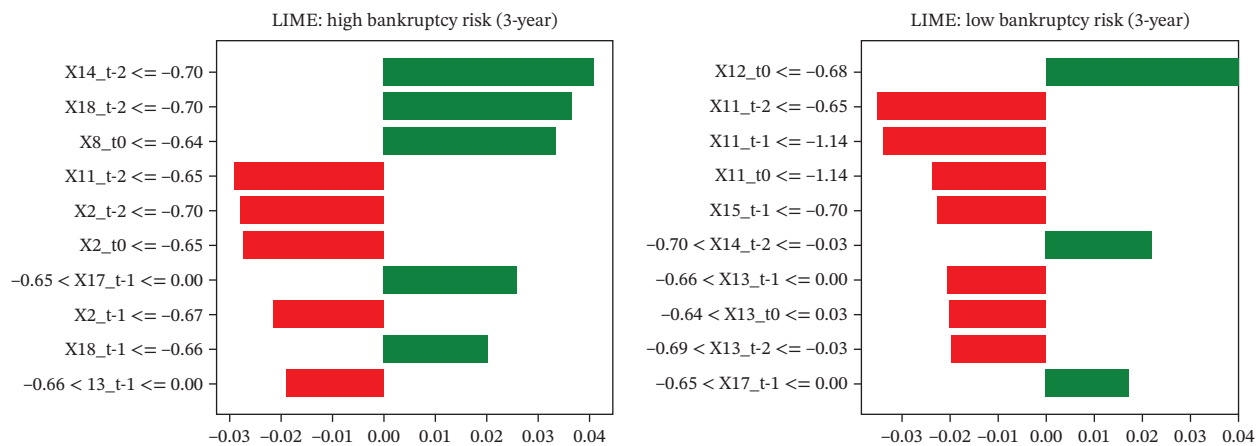


FIGURE 6 | Local interpretable model-agnostic explanations (LIME) identifying critical positive and negative feature contributions for specific firms with high and low bankruptcy risks.

with qualitative signals to align the framework with the current multimodal SOTA.

6.1 | Influence of Dataset Bias and Horizon Stability

Furthermore, the remarkable stability of the GST-Net's performance across the 1-to-3-year horizons highlights both the efficacy of the proposed architecture and the inherent characteristics of the underlying dataset. The dataset focuses on publicly traded US companies listed on the NYSE and NASDAQ, which introduces a natural selection bias toward mature, heavily regulated firms. In major public equity markets, corporate insolvencies are rarely sudden; they typically manifest as protracted periods of structural decay. Consequently, clear predictive signals, such as sustained margin compression and systemic debt accumulation, are often present in the data multiple years prior to a formal Chapter 11 filing. While the GST-Net's gating mechanism successfully isolates these long-term structural trends from short-term market noise, it is important to recognize that this prolonged visibility of distress is partly an artifact of public market bias. Future research applying this framework to private enterprises or small-cap OTC markets, where failure events are often more abrupt and less heavily telegraphed, will be essential to fully map the boundaries of the model's long-horizon stability.

Future research will integrate unstructured textual data into the pipeline. Natural language processing can extract sentiment from quarterly earnings reports. The multimodal learning strategies are expected to combine financial ratios with a text signal. Graph artificial intelligence has the ability of modeling complex interdependency in supply chain networks. Generalization to the European market is a test of global generalization by which the dataset is expanded. There are the macroeconomic indicators which can provide additional accuracy in long-term forecasting in the interest rate. High-frequency predictive modeling may become feasible with the integration of real-time market data. Self-supervised learning strategies are potentially effective in minimizing the need to use labeled data. Transformer networks can be used to compute longer historical sequences that can be used to add context. There will be dynamic graph structures that will represent changing relationships between financial entities.

Financial noise adversarial training schemes would also be more robust to financial noise. Uncertainty quantification of forecasts is possibly better optimized by Bayesian neural networks. The cross-domain adaptation will be promising to transfer the knowledge to the various industries. Federated learning can maintain privacy and consolidate data in corporations. Such developments will end up creating more robust financial monitoring systems.

While the proposed GST-Net incorporates a dual-branch architecture and a three-model ensemble strategy that introduces greater computational complexity than conventional baseline models, this additional complexity is empirically justified by its superior predictive performance. To provide a transparent assessment of computational efficiency, the execution latency of all competing models was benchmarked, and the results are presented in Table 6. Lightweight, highly optimized tree-based approaches such as XGBoost complete inference on the test cohort in approximately 0.03 s. In comparison, GST-Net requires between 0.37 and 0.49 s to process the entire test set. However, in the context of corporate solvency auditing, prediction tasks are typically performed on a quarterly or annual basis rather than in real-time or latency-critical environments. As a result, this subsecond inference time has a negligible impact on practical deployment. Moreover, the modest increase in computational cost and execution time is substantially offset by the model's remarkable improvement in minority-class detection, achieving a MCC of 0.5704 compared with only 0.3294 for the fastest baseline models. These findings demonstrate that the additional architectural complexity of GST-Net represents a practical and necessary design choice rather than an unnecessary engineering burden, as it significantly reduces the risk of costly corporate default misclassification and delivers substantially greater value in real-world financial risk assessment.

7 | Conclusion

This study establishes the GST-Net as a robust solution for the persistent challenges in corporate insolvency forecasting. By integrating a learnable gating mechanism with a dual-branch architecture, the framework successfully filters input noise while simultaneously modeling local volatility and global strategic trends. This hybrid approach effectively addresses the

performance degradation typical of standard deep learning baselines when forecasting over extended periods. Empirical results confirm the model's stability, particularly for the challenging three-year prediction horizon, where it achieved an MCC of 0.5704, a macro F1 score of 78.38%, and an accuracy of 95.95%. When evaluated against comprehensively hyperparameter-tuned tabular and deep learning baselines, GST-Net nearly doubles the minority-class detection capability of the strongest competitor (XGBoost, MCC 0.3294). Importantly, despite its architectural complexity, the framework maintains a highly practical sub-second inference time, proving that its computational overhead is both operationally negligible and strictly necessary for robust risk mitigation. The implementation of an asymmetric loss function further resolved critical class imbalance issues, ensuring that the model remains sensitive to rare bankruptcy events without sacrificing overall precision.

Beyond predictive accuracy, the framework bridges the gap between complex neural architectures and regulatory requirements through the integration of SHAP and LIME interpretability protocols. These tools provide financial stakeholders with transparent, firm-level risk attributions, effectively capturing the dynamic shift from acute, short-term liquidity crises to long-term structural solvency decay across different forecasting horizons. While the current system relies on quantitative accounting ratios, future investigations will expand the predictive pipeline to include unstructured textual disclosures and macroeconomic indicators. Subsequent research will also focus on testing the architecture against European market data and exploring graph neural networks to capture the complex, evolving interdependencies between supply chains and financial entities.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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